REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 3 Lesson 3 | Free Fall

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Take upward as positive unless specified: $a = -g$; $g = 9.8\ m/s^2$; $v_f = v_i - gt$; $\Delta y = v_i t - \frac{1}{2}gt^2$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 71–76.

P01. Free fall means motion under:

A. No force at all B. Only air resistance
C. A constant upward force D. Gravity as the only significant force

Answer

D. Gravity as the only significant force.

P02. At the highest point of an ideal vertical throw, acceleration is:

A. Downward with magnitude g B. Zero
C. Upward with magnitude g D. Dependent on the ball's speed

Answer

A. Downward with magnitude g.

P03. Neglecting air resistance, objects of different masses near Earth have:

A. Acceleration proportional to mass B. The same free-fall acceleration
C. Zero acceleration D. Different acceleration solely because of mass

Answer

B. The same free-fall acceleration.

P04. With upward positive, free-fall acceleration is:

A. $+9.8\ m/s^2$ always B. 0 at the top
C. $-9.8\ m/s^2$ D. $-9.8\ m/s$

Answer

C. $-9.8\ m/s^2$.

P05. When an ideal dropped object falls, its speed:

A. Remains zero B. Decreases uniformly
C. Depends only on its colour D. Increases uniformly with time

Answer

D. Increases uniformly with time.

P06. An object is dropped from rest. Neglect air resistance and use downward as positive. Find its velocity after 1 s. Use $g = 9.8\ m/s^2$.

A. 19.6 m/s B. 4.9 m/s
C. 9.8 m/s D. -9.8 m/s

Answer

C. 9.8 m/s. $v = gt = 9.8 \times 1 = +9.8$ m/s downward.

P07. An object falls from rest for 1 s. Neglect air resistance. Find the distance fallen using $g = 9.8\ m/s^2$.

A. 9.8 m B. 2.45 m
C. -4.9 m D. 4.9 m

Answer

D. 4.9 m. $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 9.8 \times 1^2 = 4.9$ m.

P08. A ball is thrown vertically upward at 19.6 m/s. Neglect air resistance. Find the time to its highest point using $g = 9.8\ m/s^2$.

A. 2 s B. 4 s
C. 1 s D. -2 s

Answer

A. 2 s. At the top $v_f = 0$. From $0 = v_i - gt$, $t = 19.6/9.8 = 2$ s.

P09. A ball is thrown upward at 19.6 m/s. Find its maximum rise. Neglect air resistance; $g = 9.8\ m/s^2$.

A. 39.2 m B. 19.6 m
C. 9.8 m D. -19.6 m

Answer

B. 19.6 m. $0 = v_i^2 - 2gh$, so $h = 19.6^2/(2 \times 9.8) = 19.6$ m.

P10. A stone is dropped from height 19.6 m. Calculate its impact speed. Neglect air resistance; $g = 9.8\ m/s^2$.

A. 39.2 m/s B. 9.8 m/s
C. 19.6 m/s D. -19.6 m/s

Answer

C. 19.6 m/s. $v^2 = 2gh$; $v = \sqrt{(2 \times 9.8 \times 19.6)} = 19.6$ m/s.

P11. Define free-fall acceleration.

Answer

The acceleration produced by gravity alone; near Earth's surface its magnitude is approximately 9.8 m/s^2 downward.

P12. Why is a falling feather not usually a good example of ideal free fall in air?

Answer

Air resistance is often significant compared with its weight, so gravity is not the only significant force.

P13. Describe velocity and acceleration while a thrown ball rises.

Answer

Velocity is upward and decreases in magnitude; acceleration remains downward at approximately g.

P14. Describe velocity and acceleration after the highest point.

Answer

Velocity is downward and grows in magnitude; acceleration remains downward at approximately g.

P15. For return to the launch height, compare ideal upward and downward speeds.

Answer

Their magnitudes are equal at the same height, but their directions are opposite.

P16. An object is dropped from rest. Neglect air resistance and use downward as positive. Find its velocity after 1.5 s. Use g = 9.8 m/s².

Answer

$v = gt = 9.8 \times 1.5 = +14.7$ m/s downward.

P17. An object is dropped from rest. Neglect air resistance and use downward as positive. Find its velocity after 2 s. Use g = 9.8 m/s².

Answer

$v = gt = 9.8 \times 2 = +19.6$ m/s downward.

P18. An object falls from rest for 1.5 s. Neglect air resistance. Find the distance fallen using g = 9.8 m/s².

Answer

$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 9.8 \times 1.5^2 = 11.03$ m.

P19. An object falls from rest for 2 s. Neglect air resistance. Find the distance fallen using g = 9.8 m/s².

Answer

$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 9.8 \times 2^2 = 19.6$ m.

P20. A ball is thrown vertically upward at 29.4 m/s. Neglect air resistance. Find the time to its highest point using g = 9.8 m/s².

Answer

At the top $v_f = 0$. From $0 = v_i - gt$, $t = 29.4/9.8 = 3$ s.

P21. A ball is thrown vertically upward at 39.2 m/s. Neglect air resistance. Find the time to its highest point using g = 9.8 m/s².

Answer

At the top $v_f = 0$. From $0 = v_i - gt$, $t = 39.2/9.8 = 4$ s.

P22. A ball is thrown upward at 29.4 m/s. Find its maximum rise. Neglect air resistance; g = 9.8 m/s².

Answer

$0 = v_i^2 - 2gh$, so $h = 29.4^2/(2 \times 9.8) = 44.1$ m.

P23. A ball is thrown upward at 39.2 m/s. Find its maximum rise. Neglect air resistance; g = 9.8 m/s².

Answer

$0 = v_i^2 - 2gh$, so $h = 39.2^2/(2 \times 9.8) = 78.4$ m.

P24. A stone is dropped from height 44.1 m. Calculate its impact speed. Neglect air resistance; g = 9.8 m/s².

Answer

$v^2 = 2gh$; $v = \sqrt{(2 \times 9.8 \times 44.1)} = 29.4$ m/s.

P25. A stone is dropped from height 78.4 m. Calculate its impact speed. Neglect air resistance; g = 9.8 m/s².

Answer

$v^2 = 2gh$; $v = \sqrt{(2 \times 9.8 \times 78.4)} = 39.2$ m/s.

P26. A ball is thrown upward at 19.6 m/s. Neglect air resistance. Find time to the top and maximum rise.

Answer

$t = v_i/g = 19.6/9.8 = 2.0$ s. Rise $= v_i^2/(2g) = 19.6$ m.

P27. A student says gravity switches off at the highest point. Explain the error.

Answer

Only the instantaneous vertical velocity is zero there. Gravity still acts, so acceleration remains downward and the ball starts descending.

P28. Two balls of equal size but different masses are dropped in a vacuum. Predict their motion.

Answer

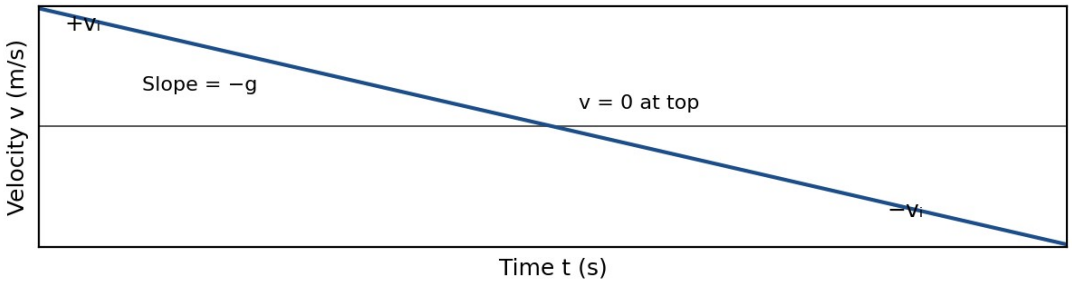
They have the same downward acceleration and, with equal initial conditions, the same velocity and position at each time. They reach the ground together.

P29. Sketch a velocity–time graph for a ball thrown upward, taking upward as positive.

Answer

Draw a straight line starting at positive initial velocity with slope −g, crossing v = 0 at the top and becoming negative during descent.

P30. Explain why equal times during a drop do not correspond to equal distances fallen.



Answer

Speed increases during free fall, so later equal time intervals cover larger distances. From rest, distance is proportional to t^2.